

STRATEGIC MARKETS MONITOR

Global Copper Balances & the Physical-Tonne Framework

Where Does the Next Tonne Come From?

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Separating physical supply constraint from demand narrative

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REPORT OBJECTIVE

To provide investors, industrial consumers and policymakers with a disciplined framework for identifying where the next tonne of copper can realistically come from, and evaluating whether future mine supply, secondary recovery and processing capacity are sufficient to meet long-term structural demand.

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About This Framework

Who Should Read This Report & Why It Matters

Copper has evolved from a traditional industrial metal into one of the world's most strategically important resources. It sits at the center of electrification, renewable energy deployment, grid modernization, artificial intelligence infrastructure, data centers, and broader industrial transformation. As economies accelerate their energy transition, the defining question is no longer whether copper demand will grow, but where the next tonne of copper will come from, and what that means for investors, industries, and policymakers.

Unlike many commodities, copper supply cannot be expanded quickly. New mines require years of exploration, permitting, financing, and infrastructure development, while declining ore grades, concentrated production, processing bottlenecks, and geopolitical risks continue to constrain future supply. The challenge facing global markets is therefore not one of demand growth, but of whether the physical supply chain can realistically deliver the volumes required to support long-term structural trends.

This report examines copper through that strategic lens. Rather than focusing on short-term price movements, it analyzes the physical foundations of the market including mine production, secondary supply, refining capacity, project pipelines, supply concentration, and the structural constraints that will shape future availability. The objective is to understand not only where future copper supply is likely to originate, but also how supply risks propagate across the value chain and influence investment decisions, industrial competitiveness, and public policy.

The analysis is intended for institutional investors, equity research professionals, commodity and macro strategists, corporate strategy teams, consulting firms, industrial companies, and policymakers seeking a deeper understanding of the forces reshaping the copper market. For investors, it provides a framework for assessing whether long-term demand expectations are supported by realistic supply growth and where structural constraints may create investment opportunities or risks. For corporations and industrial consumers, it highlights the implications for procurement strategies, capital allocation, supply-chain resilience, and strategic partnerships. For governments, sovereign wealth funds, and advisory professionals, it offers insights into resource security, critical minerals policy, industrial competitiveness, and the geopolitical implications of an increasingly constrained copper market.

Ultimately, this Copper Monitor moves beyond conventional commodity analysis to examine the intersection of geology, resource availability, industrial demand, geopolitics, and capital allocation. Its central premise is simple: the next copper cycle will be determined not only by how much copper the world requires, but by whether the world can realistically deliver the next tonne.

INSTITUTIONAL INSIGHT

Copper Is Becoming Strategic Infrastructure

Copper's role in the global economy is changing in kind, not just in scale. It sits beneath electrification, artificial-intelligence setups, data-center buildout, transmission and distribution networks, electric vehicles, industrial automation and renewable-energy generation. Few other critical minerals have exposure this broad as most substitute into a narrower set of end-uses, whereas copper is close to a common input across almost every energy-transition technology.

This breadth is changing the nature of demand itself. A growing share of copper consumption is now anchored to policy commitments including grid build-out targets, electrification mandates, data-center power capacity rather than to the general business cycle. Demand that is policy-anchored moves more slowly than GDP, but it is also less prone to reversal.

Because copper is installed once and used for decades in wiring, motors, transformers and grids, it increasingly behaves like long-duration infrastructure capital rather than a discretionary industrial input whose consumption tracks output.

Strategic Relevance: Investors should increasingly analyze copper as infrastructure supporting long-duration capital investment rather than as a traditional cyclical commodity.

1. Executive Summary

The copper debate is usually framed around price. The more consequential question for capital allocation and industrial policy is physical: Which tonne of copper arrives next whether mined, recycled, or freed by demand restraint, and what must be true for it to arrive on schedule. Our framework's central finding is that today's supply is adequate, but the forward pipeline is not, and that gap, not the price tape, is what should be tracked. Five key findings frame the rest of this report.

1. The pipeline, not the ore body, is the binding constraint. World mine production rose from about 13.6 Mt (2002) to 23.0 Mt (2024), implying supply is not scarce today. But the committed project pipeline covers only about 69% of the 2035 primary-supply requirement of roughly 25.3 Mt, leaving a gap near 7.8 Mt per year, about 31% short. ***Why it matters:** Capital committed today, not reserves in the ground, sets the effective ceiling on 2035 supply.*

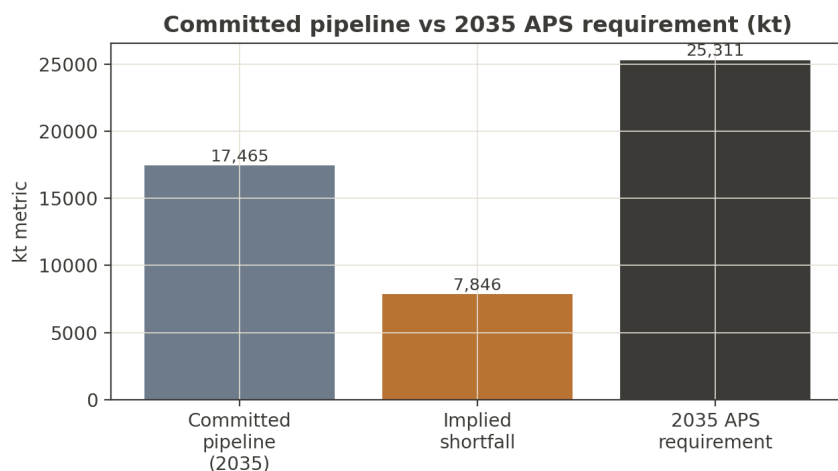
2. Refining concentration, not mining concentration, is where near-term stress sits. Mining is moderately concentrated (top-3 producers ~49.9% of world output). Refining is materially more concentrated (top-3 ~61%), and China alone smelts about 49% of world refined copper on only about 8% of mined concentrate. ***Why it matters:** Treatment and refining-charge volatility, and single-country processing risk, are downstream phenomena that a mine-only view of supply will not capture.*

3. Recycling scales fastest, but cannot close the gap alone. The secondary supply and reuse roughly doubles, from 4.1 Mt (2021) to 10.0 Mt (2040), lifting the secondary share of demand from about 17% to 27.5%. Even doubled, it leaves a primary-supply requirement that plateaus near 25 Mt. ***Why it matters:** Scrap policy is the fastest lever available to industry and government alike, however, it buys time against the pipeline gap, not closure of it.*

4. Electrification is adding a structural, largely non-cyclical demand premium. The electrification premium demand attributable to clean-technology growth beyond a structural trend rises from about 6.6 Mt (2030) to 11.0 Mt (2040), tracking the growth in cleantech demand itself. ***Why it matters:** Grid- and EV-driven demand is policy-anchored rather than cyclical, which changes how it should be modelled relative to traditional, construction-linked copper use.*

5. Raising climate ambition from a pledges-based to a net-zero pathway widens the gap further. Under a net-zero trajectory, total demand reaches roughly 33 Mt by 2030 and about 39 Mt by 2040 layering an estimated 1.9–2.6 Mt of additional requirement onto a pipeline already about 31% short of 2035 needs. ***Why it matters:** A net-zero pathway does not just require today's pipeline gap to close rather it requires mines and recyclers to mobilize faster than a pledges-based pathway already assumes.*

Figure 1. How much of the copper story is pipeline, not price?



Strategic relevance: The committed pipeline covers roughly 69% of the 2035 primary-supply requirement, leaving an explicit, quantified gap of about 7.8 Mt per year that scrap, brownfield expansion, and greenfield capacity must jointly close. *This single relationship: committed capacity versus requirement is the primary fact behind every other finding in this report.*

Source: Committed-project pipeline data, cross-referenced against IEA primary-supply milestones.

KEY TAKEAWAY

Today's copper market is not short of ore rather it is short of committed capital. The 2035 primary-supply requirement of roughly 25.3 Mt is covered only to about 69% by the committed pipeline. That residual, not the spot price, is the number to track since it tells decision makers whether the next tonne is already accounted for, or still has to be found.

What decision makers should watch: The pace at which announced projects convert to committed status, the trajectory of secondary supply's share of demand, and any shift in climate-policy ambition from a pledges-based to a net-zero trajectory each of which moves the size of the gap this report quantifies.

2. What Makes This Framework Different?

Any framework claiming to add value over conventional copper coverage should be able to say, precisely, what it does differently and why that difference changes a decision rather than merely a commentary. This section makes that case before the report turns to evidence.

Most copper research is organized around price including where it is headed, and the underlying reasons. And, whilst that framing is useful for trading desks, it is less useful for the capital-allocation and industrial-policy decisions that this report is written for. Those decisions turn on physical tonnes including where they are produced, where they bottleneck, and how fast each source can respond.

EXHIBIT 1.

What Makes This Framework Different?

Moving beyond commodity price forecasts to a physical-tonnage framework.

1. CONVENTIONAL VIEW

Traditional Copper Analysis

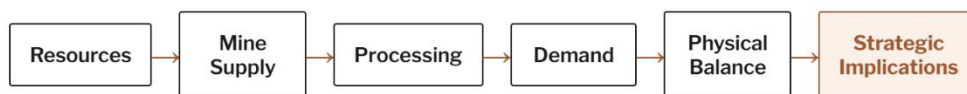
- Price
- Supply deficits
- China demand
- EV demand
- Market sentiment

2. OUR APPROACH

This Framework

- ✓ Physical tonnes
- ✓ Secondary supply
- ✓ Committed pipeline
- ✓ Processing bottlenecks
- ✓ Refining concentration
- ✓ Scrap elasticity
- ✓ Strategic resilience
- ✓ **Where does the next tonne come from?**

3. FRAMEWORK FLOW



The objective is not to predict copper prices.

It is to explain where future physical supply can realistically originate.

These distinctions are not academic. Applied consistently, they generate eight key messages that recur throughout this report, set out below and referenced by figure and table wherever the underlying evidence is presented in full.

3. Key Messages

“The market is short of committed capital, not of ore.” World mine production nearly doubled between 2002 and 2022, yet the committed pipeline covers only ~69% of the 2035 primary-supply requirement implying that the gap is a financing and permitting story, not a geological one. **Implication:** Capital-allocation and permitting decisions taken over the next one to two years, not incremental exploration, are what will determine whether 2035 supply clears the requirement.

“The choke point has moved downstream.” Refining concentration (top-3 ~61%) now exceeds mining concentration (~46%); China alone smelts ~49% of world refined copper on ~8% of mined concentrate. **Implication:** A strategy that diversifies mining jurisdictions without a parallel refining strategy leaves the system's binding constraint unaddressed.

“Recycling is the fastest lever, not the whole answer.” Secondary supply roughly doubles by 2040 and lifts its share of demand from ~17% to ~27.5%; meaningful, but insufficient to close a primary-supply requirement that still plateaus near 25 Mt. **Implication:** Scrap-collection policy deserves urgent attention, but should be sized and communicated as a gap-narrowing measure, not a gap-closing one.

“Electrification is a floor, not a cycle.” The electrification premium meaning copper demand attributable to the energy transition alone rises from ~6.6 Mt (2030) to ~11.0 Mt (2040), a structurally different demand driver than traditional, construction-linked use. **Implication:** models that treat grid- and EV-linked copper demand as cyclical will understate its persistence through a downturn.

“Higher climate ambition does not change the ordering of solutions but it lengthens the queue.” Moving from a pledges-based to a net-zero trajectory adds 1.9–2.6 Mt of extra requirement on top of a pipeline already ~31% short. **Implication:** Policymakers accelerating decarbonization timelines should expect to accelerate permitting and recycling-collection policy in parallel, not in sequence.

“Concentration risk is now a downstream risk as much as an upstream one.” Chile, Peru and the DRC dominate mining; China dominates refining where a disruption in either tier propagates quickly given the absence of comparably sized alternatives. **Implication:** Supply-chain resilience planning that stops at mine-site diversification addresses only half of the system's concentration risk.

“Water stress, not resource scarcity, is the more likely operational disruptor.” About 52% of copper mines sit in high water-stress areas, concentrated in the same three countries that dominate mined supply. **Implication:** Water-stress mitigation in Chile, Peru and the DRC is now a supply-security question for industrial buyers and governments alike, not solely an ESG one.

“This is a discipline, not a forecast.” The framework's purpose is to flag where committed capacity, secondary supply and demand ambition stand relative to the physical requirement, refreshed as production, scenario and reserve data update. **Implication:** the framework is designed to be re-run each quarter as new data arrive, not re-written each time the copper price moves.

4. Why Copper Must Be Read in Tonnes, Not Dollars

A physical system, not a price tape

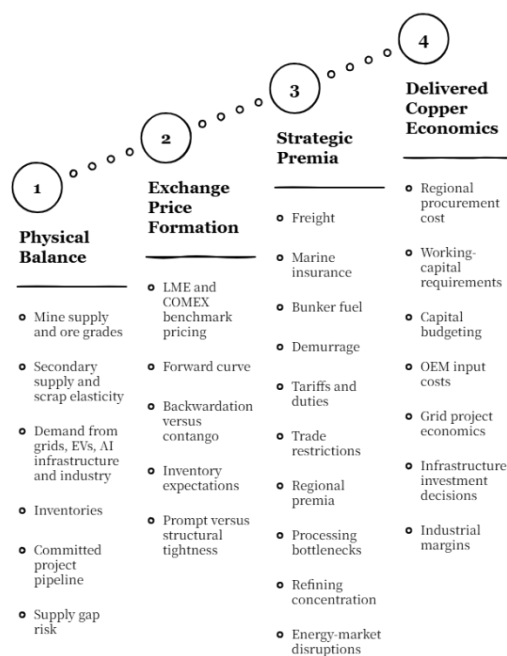
Every section of this report answers a version of the same question. Does today's copper system generate enough physical tonnes, on schedule, to meet a specific requirement? This section addresses the most basic version of that question including the unit of account itself, precisely because a framework built in dollars and one built in tonnes will not agree on when a shortfall exists.

This report works throughout in metric tonnes of contained or refined copper rather than in dollars. Price is a signal that clears a market after the fact; it does not tell a capital allocator, a refiner, or a grid planner how many physical tonnes will be available in 2030 or 2035, or from which of several structurally different sources those tonnes must come.

Demand in this framework has been split by geography including China and the rest of the world, and by end-use such as electricity grids, electric vehicles, other clean technologies, and traditional uses. Supply is split into operating mines, committed projects, and secondary (scrap) flows, then filtered through smelting and refining capacity, since a tonne of concentrate that cannot reach a smelter is not a tonne of usable copper.

Figure 2. Copper pricing framework

Strategic Framework: From Physical Balance to Delivered Copper Cost



The copper price observed on the LME or COMEX is only the starting point for industrial consumers. The final procurement cost increasingly reflects a sequence of physical, logistical and geopolitical adjustments layered onto the exchange price. Mine supply, inventories and committed capacity establish the underlying market balance, while freight, energy costs, insurance, tariffs and regional premia determine the delivered economics faced by manufacturers and infrastructure developers.

Moreover, unlike traditional commodity cycles, today's copper market operates within a fragmented geopolitical environment where disruptions to maritime trade, refining capacity or energy logistics can materially increase delivered costs without requiring a corresponding deterioration in mine supply fundamentals. Consequently, identical LME prices can translate into meaningfully different procurement costs across regions.

STRATEGIC TAKEAWAY

Distinguishing between structural physical tightness and temporary logistics or geopolitical premia helps investors separate long-term repricing from short-term market dislocations. This distinction is increasingly relevant when evaluating miners, refiners, industrial manufacturers and companies whose profitability depends on reliable access to copper rather than just its quoted exchange price.

5. The Framework: Separating What Is Mined From What Is Required

Purpose

The framework's discipline is to separate two things usually blurred together in a single copper narrative: how much copper the world is on track to produce, and how much it is on track to need. The conversion of copper from a single demand story into a dual step answer that states what the mines and scrap streams say, as well as what demand ambition requires, is what allows the gap between them to be tracked and re-estimated as new data arrives.

Scenario logic

- A pledges-based pathway: published milestones for total demand, cleantech demand, secondary supply, and primary supply requirement for near-term, mid-decade and 2040 horizons; the mid-decade value used throughout is a straight-line interpolation of the two published milestones either side of it.
- A structural base case: a counterfactual built specifically for this report: traditional uses grow at their pledges-based trend, while cleantech demand is frozen at its most recent observed level. This isolates the copper attributable specifically to the energy transition.
- A net-zero pathway: a higher-ambition trajectory drawn from the same institutional source, transcribed from published text and charts rather than a milestone table; its primary-supply requirement is inferred as total demand minus secondary supply, since no net-zero primary-supply milestone is separately published.

Reading this report's numbers with appropriate caution

Some words of caution: Not every number in this report carries the same evidentiary weight, and the report is written to make that visible rather than to flatten it. Some figures are observed directly from published production, reserve, and demand tables. Others are interpolated between two published milestones. A further set are derived including those calculated within this framework from other published inputs, such as the electrification premium or the scenario deltas. A last set are best-available approximations, derived from institutional charts and commentary. The readers should treat the last category with the most caution, and the report flags it wherever it appears.

With the unit of account and the evidentiary hierarchy established, the report turns next to the framework itself to answer how mined, secondary and required tonnes are separated and tracked against one another through 2035 and 2040.

6. Supply Is Not Scarce Today — The Pipeline Is Short

World mine production rose from about 13.6 Mt in 2002 to 23.0 Mt in 2024. Supply is geographically concentrated. Chile (5.5 Mt, 2024) remains the largest producer, and the Democratic Republic of the Congo (11 approximately 3.2 Mt, 2024) has overtaken Peru (11 approximately 2.8 Mt, 2024) to become the second-largest source. Chile alone supplies about 24% of mined copper. The DRC is the fastest-growing major source, whose output rose more than six-fold between 2010 and 2022 and continued climbing through 2024. Table 1 sets out the underlying country-level production and reserve figures in full.

Figure 3. How much has world mine production grown?



Strategic caption: World mine production has nearly doubled since 2002, but growth has flattened since roughly 2019–2020.

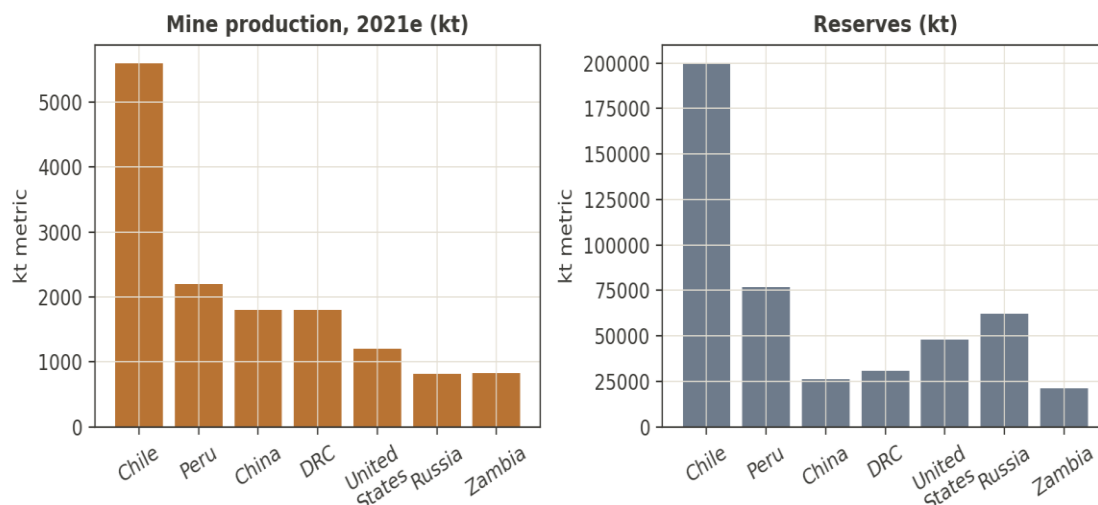
Why this matters: A flattening trend line, not a falling one, is what makes the pipeline gap consequential — growth from here depends on committed capital, not an organic extension of the historical curve.

Table 1. Country mine production and reserves

Country	Mine production (kt, latest)	Reserves (kt)
Chile	5,600	200,000
Peru	2,200	77,000
China	1,800	26,000
DRC (Congo)	1,800	31,000
United States	1,200	48,000
Russia	820	62,000
Zambia	830	21,000
World total	21,000	880,000

Strategic interpretation: Reserve depth does not track current output one-for-one as several major reserve holders are not proportionally large current producers, which is precisely why the committed pipeline, not the reserve base, is the binding near-term constraint.

Source: ICSG world mine production series, 2002–2022.

Figure 4. Where does production sit relative to reserves?

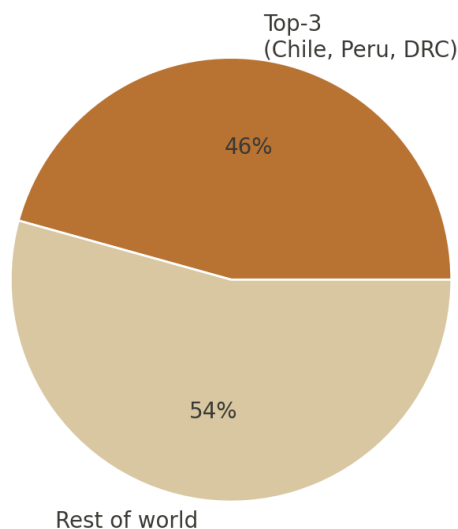
Strategic caption: Reserve depth is not evenly distributed relative to current production — Chile and Peru hold reserves far in excess of current output, while several mid-tier producers have thinner cushions.

Why this matters: Reserves are a ceiling on what could be mined, not a schedule for what will be — the gap between the two panels is where permitting and investment timelines matter most.

Source: USGS Mineral Commodity Summaries, country production and reserves.

Figure 5. How concentrated is mining, really?

Mining concentration, top-3 vs rest of world (2021)



Strategic caption: Chile, DRC and Peru together account for roughly 50% of world mined copper (2024, approx.), up from 46% in 2021.

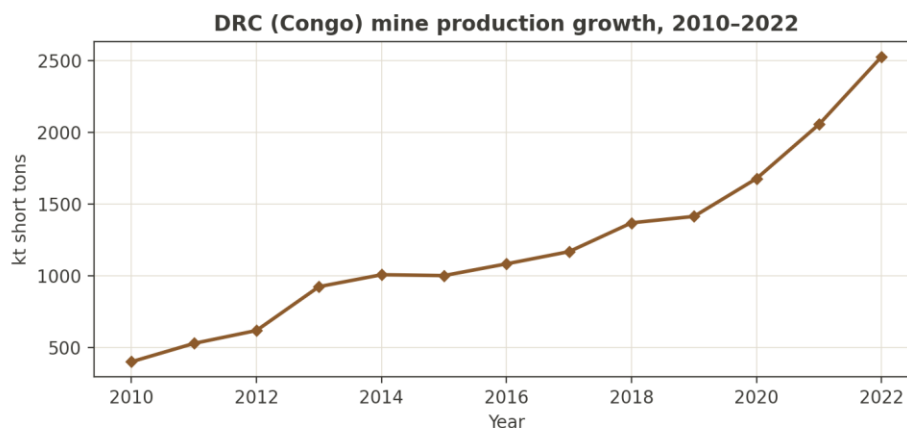
Why this matters: A single-country disruption in any of the top three — flooding, seismic events, permitting delays — propagates quickly into the global balance because there is no comparably sized fourth producer to absorb it.

Source: Country-level mine production, top-3 cross-check against pledges-based concentration series.

The DRC's trajectory, shown in Figure 6 below illustrates of how quickly a producer can move up the concentration ranking once committed capital arrives: a six-fold increase in twelve years has made it structurally part of the top-3 group alongside Chile and Peru. For industrial-policy purposes, this is also the argument for treating critical-minerals strategy as a permitting and financing exercise rather than a geological one. The DRC's reserves were not newly discovered over that period; the capital and infrastructure to convert them were newly committed. The same logic applies in reverse to resource-nationalism risk: a jurisdiction that revises fiscal or permitting terms after capital has been committed does not

just slow one project, it slows the specific mechanism, that is, conversion of reserves to committed output, which is, the real ‘binding constraint’.

Figure 6. How fast has the DRC become a top-tier producer?



Strategic caption: DRC output rose from roughly 400 kt (2010) to 2,500+ kt (2022) — more than a six-fold increase in twelve years.

Why this matters: *The DRC is now structurally part of the top-3 producer group, which raises its country-risk profile to the same strategic tier as Chile and Peru.*

Source: ICSG country-level mine production series.

INSTITUTIONAL INSIGHT

Why the Era of Easy Copper Is Ending

The marginal tonne of copper is becoming progressively harder to produce. Ore grades in mature districts such as Chile and the United States have been declining for decades, meaning more rock must be mined and processed to yield the same tonne of metal, in turn raising both capital and energy intensity per unit of output.

Much of the world's high-grade, easily permitted resource base has already been developed. The next marginal project increasingly sits in more remote, more complex and more politically exposed geographies, where permitting timelines have lengthened, water stress is a binding operational constraint, and social license to operate and resource-nationalism risk add further delay and cost.

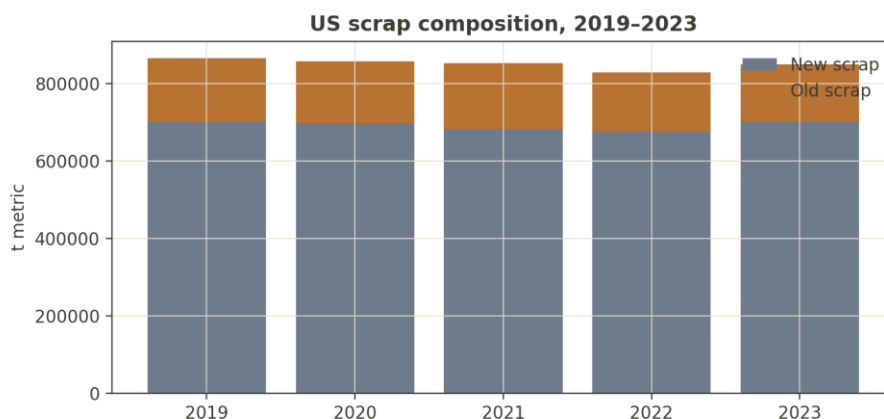
Greenfield copper projects typically take fifteen to twenty years from discovery to first production. That lag means the supply base available in 2035–2040 is already substantially determined, or foreclosed by investment decisions made, or deferred, today.

Strategic relevance: Future supply is increasingly determined by investment decisions already made rather than by prices observed today.

7. Recycling Is the Fastest Lever — Not a Sufficient One

Secondary copper is the most price-responsive supply source. In the United States, old (post-consumer) and new (manufacturing) scrap together delivered about 0.83 Mt of recycled copper in the most recent year, meeting 32–36% of US supply. Old scrap being the elastic, post-consumer stream, is the smaller share, about 0.15–0.17 Mt, which is why scrap responds to price but with a lag and a ceiling set by the stock of end-of-life metal. Additionally, the secondary-share figures used throughout this report (16.5% rising to 27.5%, Table 6, indicator 7) measure IEA APS secondary refined supply against total demand. This differs from ICSG's broader recycling input rate reported in the Factbook 2025 at roughly one-third of global copper use in 2024 which also counts direct-melt scrap bypassing the refinery entirely. The two are not directly comparable, and this report uses the narrower, demand-denominated measure throughout for consistency with the APS scenario framework.

Figure 7. What is scrap made of?

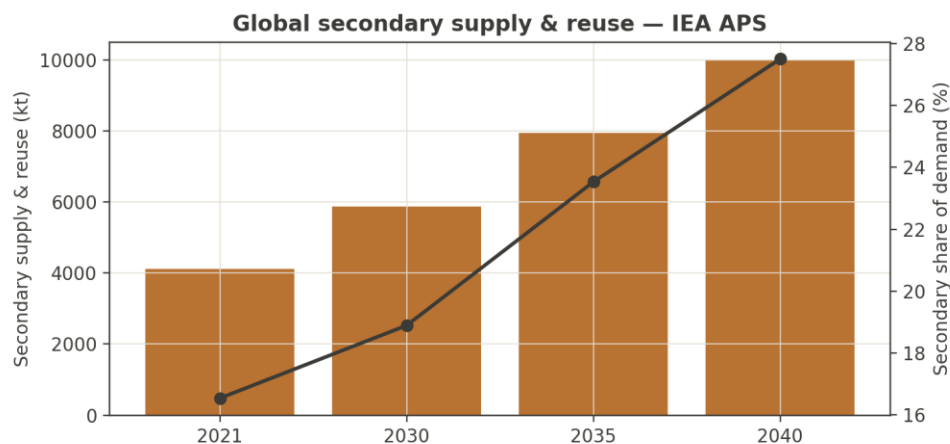


Strategic caption: New (manufacturing) scrap consistently outweighs old (post-consumer) scrap by more than 4:1.

Why this matters: Old scrap is the tonne that responds to price incentives; new scrap is closer to a fixed byproduct of current manufacturing volume, hence, a policy aimed at 'more recycling' should specify which stream it targets.

This distinction has direct relevance for processing-diversification and friend-shoring policy. Thereby, a jurisdiction seeking to build secondary-copper capacity as an alternative to mined-concentrate dependence should design collection infrastructure specifically around old, post-consumer scrap, since new scrap tracks domestic manufacturing activity and cannot be expanded by policy in the same way.

Source: USGS recycling statistics, US scrap composition.

Figure 8. How far can recycling scale?

Strategic caption: Global secondary supply and reuse roughly doubles, from 4.1 Mt to 10.0 Mt, while its share of total demand rises from about 17% to 27.5%.

Why this matters: Recycling is the fastest-scaling source of new supply in this framework — but even doubled, it leaves a primary-supply requirement near 25 Mt that mines alone must still meet.

For governments weighing strategic stockpiles against recycling investment, Figure 8 is the relevant reference point: secondary supply's near-doubling is real and worth pursuing aggressively, but it does not remove the case for a stockpile or a committed-pipeline strategy instead it only changes how large that strategy needs to be.

Source: Pledges-based secondary-supply and total-demand milestones.

STRATEGIC TAKEAWAY

Scrap policy directed towards improving collection rates and standardizing secondary-metal quality, is the single fastest lever available to industry and government alike.

INSTITUTIONAL INSIGHT

Copper Behaves Differently from Oil

Oil is consumed at the point of use. Copper, on the contrary, once installed in wiring, motors, pipes or grid infrastructure, remains within the economy for decades, building a growing above-ground “urban mine” that can eventually be recovered. This is why secondary supply is a genuine, structural growth story rather than a residual by-product stream, and why collection infrastructure and scrap quality matter as much to the copper balance as mining economics do.

But recycling cannot substitute for primary supply. Even as secondary output grows, it can only be drawn from metal that has already reached end-of-life; a growing in-use stock requires continued primary mine supply to be built up in the first place.

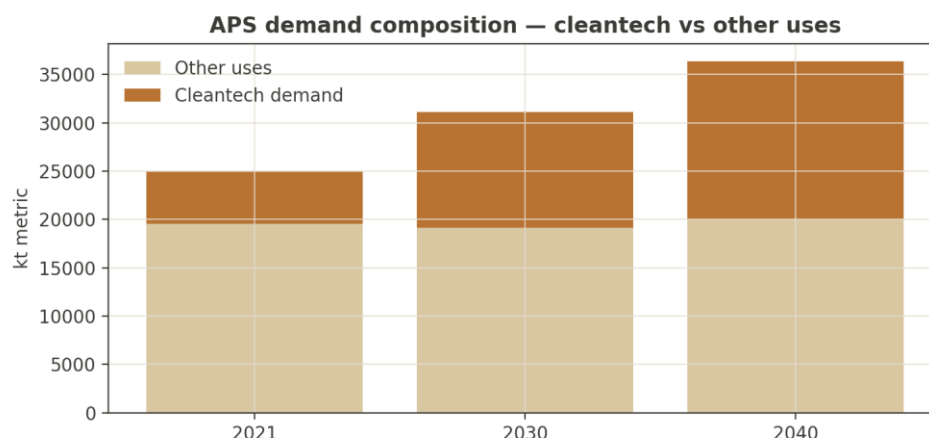
Copper also compares differently with lithium, nickel and cobalt. Those chemistries can sometimes be reformulated or substituted at the battery-cell level. Copper's combination of conductivity, ductility, durability and corrosion resistance means substitution where aluminium in some applications can reduce intensity at the margin, but cannot remove copper from the core of grids, motors and electrical infrastructure.

Strategic relevance: Copper is simultaneously recyclable and principally irreplaceable.

8. Demand Growth Alone Will Not Explain the Copper Story

Total copper demand grows from about 24.9 Mt to 31.1 Mt by 2030 and 36.4 Mt by 2040 on a pledges-based pathway. The growth is compositional as much as volumetric: clean-technology demand more than triples, from 5.4 Mt to 12.0 Mt (2030) and 16.3 Mt (2040), while traditional 'other uses' stay roughly flat near 19–20 Mt.

Figure 9. Is copper demand growth about volume or composition?

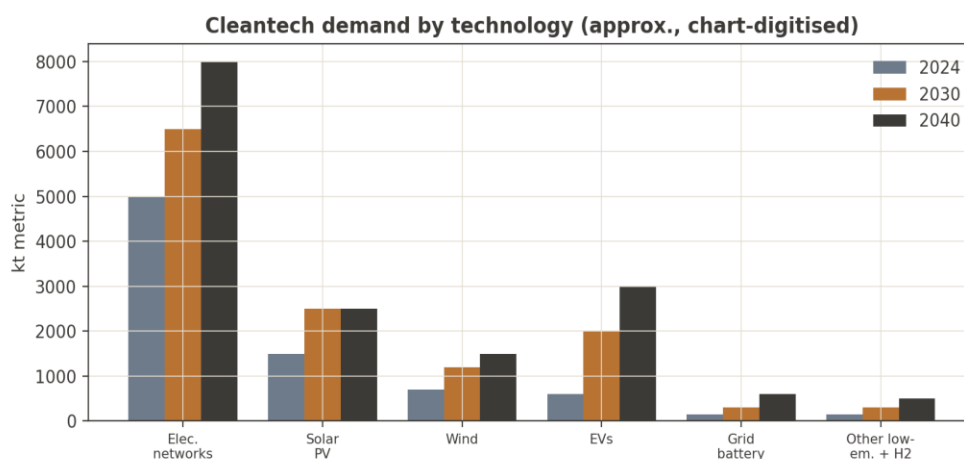


Strategic caption: Cleantech demand grows from roughly 22% of total demand to about 45% by 2040, while traditional uses are essentially flat in absolute terms.

Why this matters: This is a compositional shift in the demand base, not merely a growth story — the copper market of 2040 is structurally different from that of today, even before considering price.

Source: Pledges-based demand-by-block milestones, cleantech vs other uses.

Figure 10. Which cleantech block drives demand?



Strategic caption: Electricity networks are the largest single cleantech demand block across the period shown, ahead of solar PV, EVs, wind, grid battery storage and other low-emissions technologies combined.

Why this matters: Grid copper intensity has no engineering-thrifting offset comparable to that available in EV battery design, so this demand block is closer to a floor than the others.

This provides a deep connection demand composition and grid resilience policy since electricity-network copper intensity cannot be engineered down the way EV battery intensity can. As a result, roughly 80 million km of new network by 2040 and the associated doubling of annual grid investment to over USD 600 billion by 2030 should be treated as a firm, policy-anchored claim on future copper supply.

Source: Cleantech demand by technology, chart-derived approximation.

Two demand blocks in particular define different 'next tonnes.' Electricity grids require on the order of 80 million km of new network by 2040 equivalent to the entire existing network, with annual grid investment roughly doubling to over USD 600 billion by 2030. Electric vehicles, at roughly 60–83 kg of copper per battery vehicle and almost 14 million units sold in the most recent year (about 60% in China), already represent on the order of 1 Mt of direct annual copper demand, rising steeply even as per-vehicle intensity declines.

STRATEGIC TAKEAWAY

China sits on both sides of the ledger. It consumes over half of global refined copper and is shifting its own demand from a property-led base toward grid and EV end-uses. This is a shift that is policy-anchored and less cyclical than construction, and that changes the shape of the demand curve, not just its size. Demand composition, however, is only half of the physical balance.

9. The Bottleneck Has Moved Downstream

Mining optimism must clear a processing bottleneck before it reaches end-users. The world smelter production reached 23.3 Mt in 2024, of which China accounted for 51%, Japan 7% and Chile 4%. Moreover, refining is more concentrated than mining: top-three refined-production share is approximately 61% (China 45%, DRC 9% via SX-EW, and Chile 7%) against about 50% for mining, and China refines roughly 12.4 Mt on only about 1.8–1.9 Mt of domestic mine output, an even wider mismatch than in 2021.

EXHIBIT 2.

Copper's Constraint Has Moved Downstream

Processing capacity—not geology—is increasingly determining copper availability.

VALUE CHAIN

Mine

Ore is extracted at grades that continue to decline globally.

Constraint: Ore grade & permitting.
Implication: Longer lead times upstream.



Concentrate

Ore is milled into a shippable concentrate near the mine.

Constraint: Milling capacity & logistics.
Implication: Trade flows shape exposure.



Smelting

Concentrate is converted into blister copper at high heat.

Constraint: Treatment charge economics.
Implication: Margins squeeze independent smelters.



Refining

Blister is refined into the pure cathode that markets trade.

Constraint: Concentrated capacity, mostly in one region.
Implication: The true chokepoint in physical supply.



Fabrication

Cathode is drawn into wire, rod, and sheet products.

Constraint: Regional fabrication capacity.
Implication: Determines delivered lead times.



End Uses

Grid, construction, and EV demand pull on finished product.

Constraint: Electrification demand growth.
Implication: Sets the size of the physical gap.

FIVE STRUCTURAL INSIGHTS

China's Refining Dominance

A single country refines the majority of global cathode, concentrating a critical chokepoint outside most consuming markets.

Processing Bottlenecks

Smelting and refining capacity expand far slower than mine output, so new tonnes can stall well before reaching market.

Capital Intensity

Building new refining capacity requires large, long-horizon capital commitments that few operators are willing to make.

The Scrap Loop

Secondary copper recovery increasingly competes with primary refining for the same finite processing capacity.

Downstream Market Power

Whoever controls refining and fabrication increasingly sets terms for price, allocation, and delivery timing.

INVESTMENT IMPLICATIONS

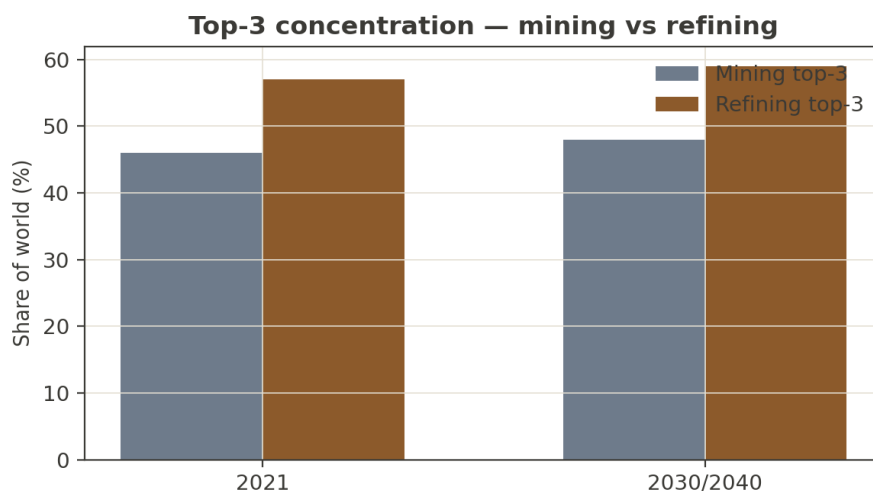
- Favor refining and fabrication capacity over pure exploration.
- Reassess exposure to single-country processing risk.

POLICY IMPLICATIONS

- Incentivize domestic and allied refining capacity.
- Support scrap-recovery infrastructure as a supply lever.

KEY INDICATORS TO MONITOR

- Treatment and refining charge trends.
- Announced smelter capacity additions and closures.

Figure 11. Which stage of the value chain is more concentrated?

Strategic caption: Refining concentration exceeds mining concentration both today and on the forward milestone, and the gap persists rather than narrowing.

Why this matters: A supply strategy that diversifies mining jurisdictions without addressing refining concentration has not removed the system's binding concentration risk — it has relocated it downstream.

The table below (table 2) quantifies the mechanism behind that relocation. The consequence is not confined to industrial buyers. For governments, a single-hub refining structure is just as much a critical-minerals-strategy question as much as a mining-policy one, since diversifying where ore is dug up does nothing to diversify where it is turned into usable metal. For investors, refining-charge volatility, implying, the treatment and refining charges smelters earn for processing concentrate, becomes a leading indicator of concentrate scarcity. This is well before it shows up in the copper price itself, since it is smelter margins, not miner margins, that compress first when concentrate is scarce relative to capacity. For industrial buyers, the practical implication is that security of supply now depends on securing refining capacity or offtake, not only mined tonnage. And for supply-chain resilience more broadly, this concentration means a single-country disruption in the form of policy, infrastructure, or otherwise in the dominant refining hub would propagate through the global balance faster than an equivalent disruption at any single mine.

Source: Top-3 concentration cross-check, mining vs refining.

This import dependency is the mechanism behind collapsing treatment and refining charges: when concentrate is scarce relative to smelting capacity, it is smelter margins and not miner margins that compress first. Secondary refined production is a partial relief valve, already about 17% of total refined output. Table 2 sets out smelter production shares by country for the most recent benchmark year.

Table 2. Smelter production shares, most recent benchmark year

Country / group	Share of world smelter production
China	49%
Japan	8%
Chile	5%
Russia	5%
Others	33%

Table 2 — Strategic interpretation: China's smelter share (51%, 2024) far exceeds its mined-concentrate share (~8%), confirming the import-dependency mechanism as the structural driver behind treatment- and refining-charge volatility. A related development the original mining-vs-refining framing did not anticipate: the DRC has become the #2 refined-copper producer (9%, 2024) entirely via SX-EW rather than smelting, adding a second, distinct route by which downstream concentration is shifting. Note on vintage: China, Japan and Chile above are 2024 figures (Factbook 2025); Russia's

5% and the 33% Others residual are carried over from the 2019 benchmark, since the Factbook 2025 text does not separately disclose a 2024 figure for Russia, they should be read as directionally indicative rather than current-year.

STRATEGIC TAKEAWAY

Building new mines without matching processing capacity does not remove the bottleneck, it relocates it. Refining diversification deserves the same strategic priority as mining diversification, and currently receives less. Even a fully resolved refining bottleneck, however, would still leave the pipeline question intact.

INVESTOR PERSPECTIVE

Copper Prices Now Contain Multiple Strategic Premia

Exchange prices for copper on the LME, COMEX and SHFE increasingly sit on top of factors well beyond mine and smelter economics: freight, marine insurance, tariffs, strategic stockpiling, financing costs and geopolitical risk premia.

The transmission mechanism runs through shipping, not through copper directly. Disruptions such as the Russia–Ukraine war, attacks in the Red Sea and Bab el-Mandeb, and tensions around the Strait of Hormuz lengthen voyages including rerouting around the Cape of Good Hope, and raise bunker-fuel demand, freight rates, war-risk insurance, demurrage and working-capital requirements across commodity supply chains broadly, well before any change occurs in physical copper availability.

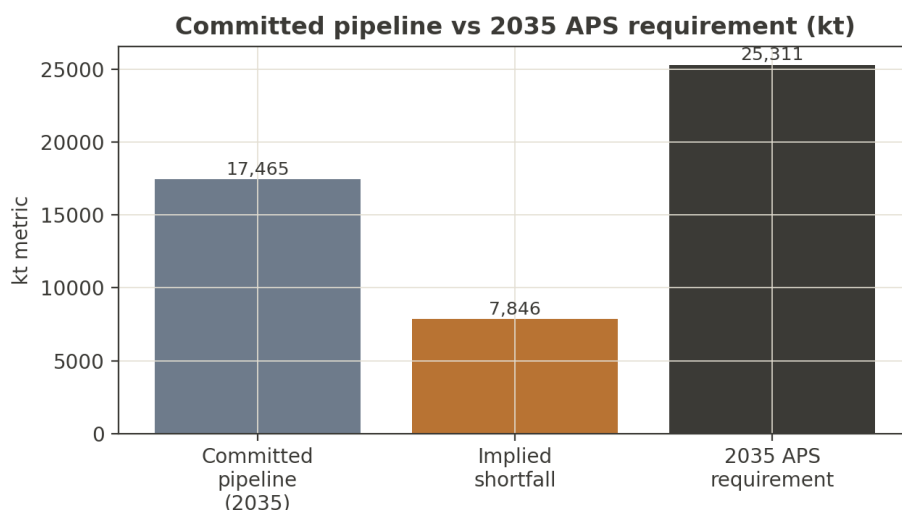
Disruptions to crude-oil shipping illustrate the same mechanism at work. That is, rerouting around Saudi Arabia's Red Sea infrastructure, such as the Yanbu terminal, or lengthening voyages around the Cape of Good Hope tightens global shipping capacity and raises freight costs across commodity supply chains generally, copper included. This is a shared logistics-capacity channel, not a claim that copper itself is transported on crude tankers.

Strategic relevance: Copper pricing increasingly reflects physical balance plus logistics, policy and geopolitical premia rather than mine economics alone.

10. Climate Ambition Raises the Bar Further

Against the mid-decade primary-supply requirement of about 25.3 Mt, the committed project pipeline is judged about 31% short. This is estimated as a gap near 7.8 Mt per year, because committed capacity of roughly 17.5 Mt covers only about 69% of the requirement. Greenfield mines, additional scrap, and demand restraint must, as a combination, fill the rest.

Figure 12. How short is the committed pipeline?



Strategic caption: The committed pipeline covers about 69% of the mid-decade requirement, leaving a quantified gap of roughly 7.8 Mt per year.

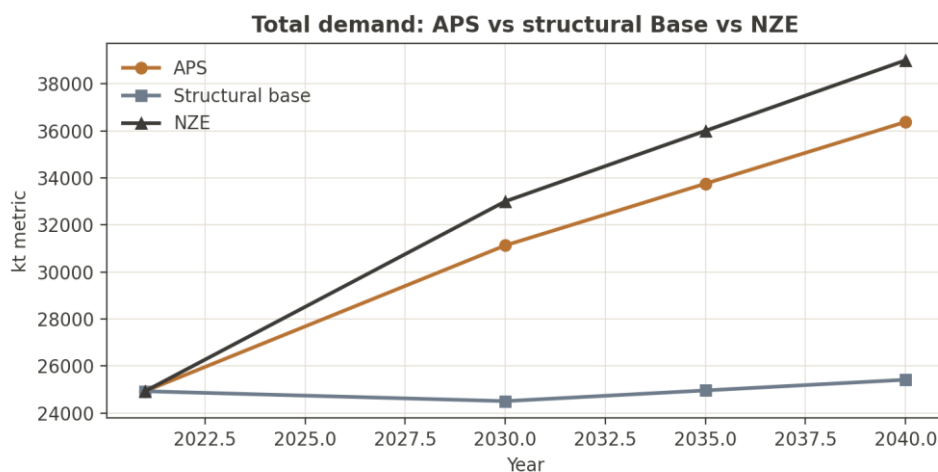
Why this matters: This is the single most load-bearing number in the report — every Strategic relevance in the sections that follow traces back to how this gap is expected to close.

For energy-security and critical-minerals-strategy purposes, Figure 12 illustrates a requirement that is already 31% uncovered before any additional climate ambition is layered on is a starting position, not a worst case. Any policy step that raises the primary-supply requirement such as faster grid build-out, accelerated EV mandates, a shift toward net zero, does so against a pipeline that has not yet closed its existing gap, which is precisely the mechanism the net-zero layer below quantifies.

Source: Committed project pipeline vs primary-supply requirement.

A structural base case for the electrification premium

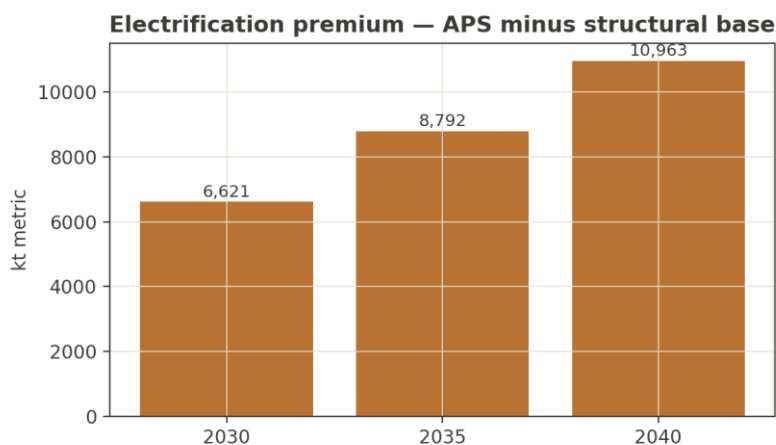
A naive extrapolation of historical demand growth exceeds the pledges-based pathway, because traditional 'other uses' plateau while cleantech triples, the total demand resultantly grows slower than the historical trend alone would suggest. The meaningful contrary to fact is a structural base case with traditional uses at trend, plus clean-technology demand frozen at its most recent level. Defined this way, the electrification premium which pledges-based demand minus the base case, rises from about 6.6 Mt (2030) to 8.8 Mt (mid-decade) and 11.0 Mt (2040).

Figure 13. How much more copper does climate ambition require?

Strategic caption: The gap between the pledges-based pathway and the structural base widens continuously through 2040; a net-zero trajectory sits further above both throughout.

Why this matters: The vertical distance between the base and pledges-based lines is, by construction, the electrification premium — the extra copper the energy transition alone requires, independent of underlying economic growth.

Source: Total demand: pledges-based pathway vs structural base vs net-zero trajectory.

Figure 14. What is electrification costing in copper?

Strategic caption: The premium nearly doubles from 6.6 Mt (2030) to 11.0 Mt (2040).

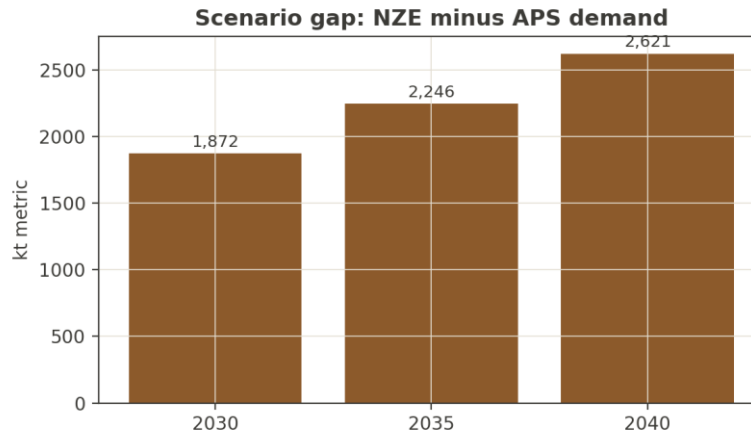
Why this matters: This premium is essentially equal to the growth in clean-technology demand itself — a useful cross-check that the base-case construction is internally consistent.

Source: Electrification premium, model calculation from published demand blocks.

A net-zero layer

Raising ambition to a net-zero trajectory adds copper on top of this. Refined demand reaches roughly 33 Mt by 2030 (versus 31 Mt on the pledges-based path) and rises to roughly 39 Mt by 2040 about 1.9 Mt more copper than the pledges-based path in 2030 and about 2.6 Mt more by 2040, and 8.5–13.6 Mt above the structural base case. Because no net-zero primary-supply milestone is separately published, this framework infers net-zero mining requirements as net-zero demand minus secondary supply, roughly 27–29 Mt per year, above the pledges-based figure of about 25 Mt.

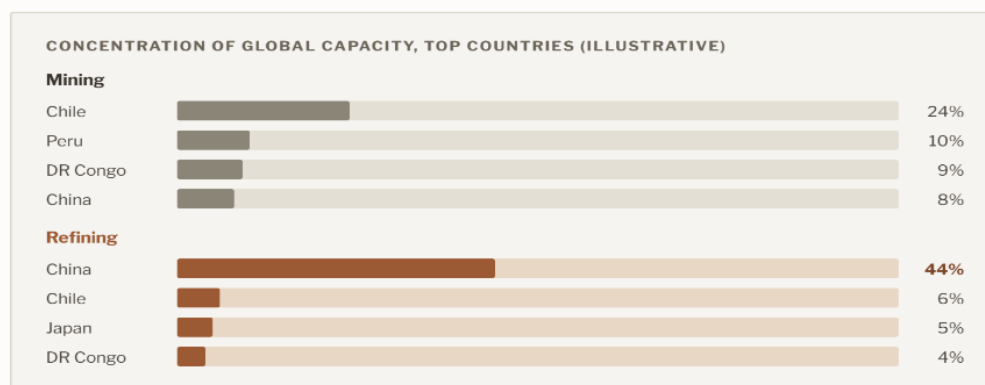
Figure 15. How much extra copper does net zero require over the pledges-based path?



Strategic caption: The net-zero-over-pledges delta grows from 1.9 Mt (2030) to 2.6 Mt (2040).

Why this matters: Raising ambition does not change the ordering of where the next tonne comes from — it lengthens the queue, demanding more of each source at once.

Source: Net-zero vs pledges-based demand delta, model calculation.

EXHIBIT 3.**Where Market Power Actually Sits***Downstream processing concentration increasingly shapes market resilience.***STRATEGIC OBSERVATIONS**

- Refining is far more concentrated than mining, reversing where real leverage sits.
- A single country's refining share exceeds any single country's mining share by a wide margin.
- Mine supply diversification does not translate into processing diversification.
- Disruption risk is now a downstream, not an upstream, story.
- New capacity announcements should be weighed against processing, not mining, capacity.

INVESTORS

Processing exposure is a distinct, underpriced risk factor separate from mine ownership.

INDUSTRY

Securing refining capacity matters as much as securing ore for reliable delivery.

POLICYMAKERS

Supply security policy must address processing concentration, not only reserves.

STRATEGIC TAKEAWAY

The duration and delivery of the committed pipeline and not the size of any single mine is what should anchor scenario planning. A net-zero pathway does not require the pipeline gap to be re-imagined, it requires the same gap to close faster.

11. Risk Register

Each risk is evaluated through its transmission channel into the report's physical-balance framework. Investment decisions should be guided by the underlying transmission mechanism and its implications for hedge ratios, capital allocation and monitoring priorities rather than by the headline event alone.

Risk Landscape at a Glance

- Skew. Risks are tilted toward a widening gap: committed-pipeline slippage and rising climate ambition both push the requirement further from committed capacity.
- Duration over discovery. The relevant question is not whether more copper exists, but how fast known projects convert from announced to committed.
- Primary signal. Committed-pipeline coverage of the mid-decade requirement is the single number that leads every other risk in this register.
- Two-sided downstream risk. Refining concentration is now as strategically significant as mining concentration.
- Chief offsetting case. Faster-than-expected secondary-supply scaling and accelerated greenfield conversion are the two paths that narrow the gap rather than widen it.

The exhibit below organizes the report's risk landscape into nine specific events, each mapped to the mechanism through which it would widen or narrow the pipeline gap.

EXHIBIT 4.

Strategic Materials Outlook

Copper Strategic Risk Register

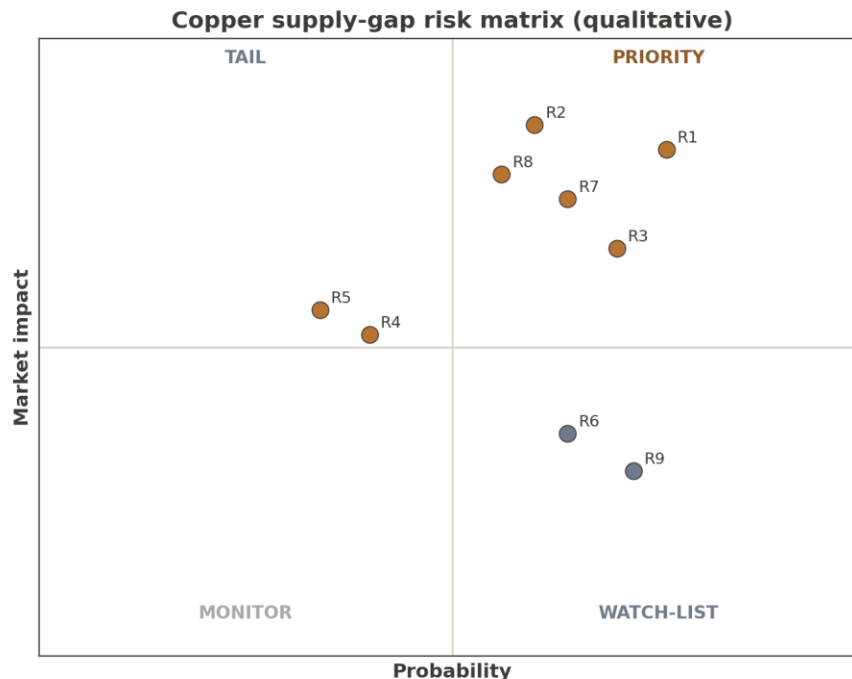
Structural risks influencing long-term copper availability.

RISK	PROB.	IMPACT	HORIZON	LEADING INDICATORS	MITIGATION	SCORE
Water stress — Chile	High	High	0–3 yrs	Rainfall deficits, aquifer permit denials, desalination approval pace	Desalination capacity buildout, brine recycling	●
Water stress — Peru	Medium	High	0–3 yrs	Community water disputes, glacier retreat rates	Community water-sharing agreements	●
Water stress — DRC	Medium	Medium	3–5 yrs	Rainfall variability, hydropower reliability	Diversify water sourcing, on-site storage	●
Permitting	High	High	0–3 yrs	Permit backlog length, litigation filings	Early community engagement, streamlined filings	●
Processing concentration	Medium	High	3–5 yrs	Smelter ownership shifts, export quota changes	Diversify smelting partners geographically	●
Refining capacity	Medium	Medium	3–5 yrs	Refinery utilization rates, capex announcements	Invest in regional refining capacity	●
Grid demand	High	Medium	5–10 yrs	Electrification targets, grid capex trends	Long-term offtake and supply contracts	●
EV demand	High	High	5–10 yrs	EV sales growth, battery chemistry shifts	Demand-side substitution research	●
Scrap availability	Medium	Medium	3–5 yrs	Collection rates, scrap price spreads	Expand urban mining infrastructure	●

● Elevated ● Watch ● Contained

EXECUTIVE OBSERVATIONS

1. Water stress across Chile, Peru, and the DRC is the most consequential near-term constraint on mined output, converging with permitting delays to compress the supply pipeline through 2029.
2. Concentration in processing and refining leaves the value chain exposed to single-geography disruption; diversification of smelting and refining capacity should be prioritized over new mine development in mitigation planning.
3. Demand-side pressure from grid buildout and EV adoption is structurally rising faster than scrap recovery can offset, sustaining a persistent tightness that favors long-term offtake contracts over spot exposure.

Figure 16. Which risks deserve priority attention versus routine monitoring?

Strategic caption: R1 and R2 pipeline slippage and rising climate ambition sit in the priority quadrant: high probability, high impact, and both widen rather than narrow the gap.

Why this matters: This qualitative triage is what should discipline monitoring effort: priority-quadrant risks warrant active hedging and capital-allocation response; watch-list risks warrant tracking but not immediate re-positioning.

Source: Qualitative positioning based on the risk register above; judgment-based, not model-derived.

The priority cluster (R1, R2, R7, R8) holds the high-probability, high-impact risks that dominate positioning; R4 and R5 are the offsetting risks that would narrow the gap rather than widen it; R6 and R9 sit in the watch-list and worth tracking but lower immediate impact.

Figure 16 translates Exhibit 3's nine risk events into a probability-impact view. When they are read together, they make a specific case for strategic stockpiles as a policy instrument: the priority-quadrant risks (R1, R2, R7, R8) are concentrated in exactly the categories, that is, committed-pipeline slippage, water stress in a three-country producer base, and single-hub refining disruption, the latter being an event where the stockpile is designed to buffer against. At the very same, the offsetting risks (R4, R5) that would narrow the gap are outside a stockpile's reach entirely and depend instead on recycling-collection policy and permitting reform.

12. Sector Strategy Matrix

Having identified which risks matter most and why, the report shifts focus from risk to response signifying what each constituent of the value chain should do for its mitigation. This section evaluates components of the copper value chain by the specific mechanism connecting the physical gap to strategic decisions, and not by a single directional copper-price view.

Upstream miners

The pipeline shortfall is a demand signal for committed capacity, specifically projects that reach committed status close the gap this report quantifies; projects that remain merely announced do not.

Refiners and smelters

Operators with secured concentrate offtake are structurally advantaged while refining stays this concentrated; new mining capacity without matching processing capacity relocates the bottleneck rather than removing it.

Scrap and recycling processors

Direct leverage on the fastest-growing supply source; policy that improves collection and standardizes secondary-metal quality expands the near-term tonne, and matters more the closer the trajectory runs toward net zero.

Industrial consumers

Grid-equipment manufacturers and other large copper consumers face a bottleneck-on-bottleneck exposure simultaneous input-cost and input-availability risk, since grid demand is close to a floor with no thrifting offset available to lessen it.

Infrastructure and grid planners

The roughly 80 million km of new grid required by 2040, and the associated doubling of annual investment to over USD 600 billion by 2030, should be planned against a primary-supply base committed to only about 69% of what the mid-2030s alone will require.

Governments and policymakers

The effective levers are permitting timelines for committed and greenfield projects, water-stress mitigation in Chile, Peru and the DRC, recycling-collection infrastructure, and diversification of refining capacity away from a single dominant hub. Each of these levers maps to a live industrial-policy debate: permitting reform and resource-nationalism risk on the mining side, friend-shoring and processing diversification on the refining side, and strategic stockpiles as a bridge while both are built out.

Table 4 summarizes the mechanism, strategic impact and key monitor for each of the six groups discussed above.

Table 4. Sector strategy matrix: transmission of the pipeline gap

Sector	Mechanism	Strategic impact	Key monitor
Upstream miners	Committed-capacity conversion closes the 2035 gap directly	Scarcity rent for de-risked, committed pipelines	Committed-pipeline capacity revisions
Refiners / smelters	Concentrate scarcity compresses smelter margins first	Advantage to secured concentrate offtake	China smelter share; treatment charges

Sector	Mechanism	Strategic impact	Key monitor
Scrap / recycling	Fastest-scaling supply source, capped by end-of-life stock	Near-term tonne expansion via collection policy	Old-scrap recycling statistics
Industrial consumers	Grid/EV demand is a floor, not a cycle	Simultaneous cost and availability exposure	Grid investment; EV sales
Infrastructure / grid	Investment need doubling against a short pipeline	Planning risk if supply assumptions overstate coverage	Grid build-out vs pipeline coverage
Governments	Permitting, water-stress and refining diversification are the effective levers	Policy timing shapes how fast the gap closes	Permitting timelines; water-stress developments

Table 4 — strategic interpretation: valuation and strategic response should follow the mechanism — upstream and committed capacity re-rate on a persistent gap; refiners are exposed on concentration, not on price alone.

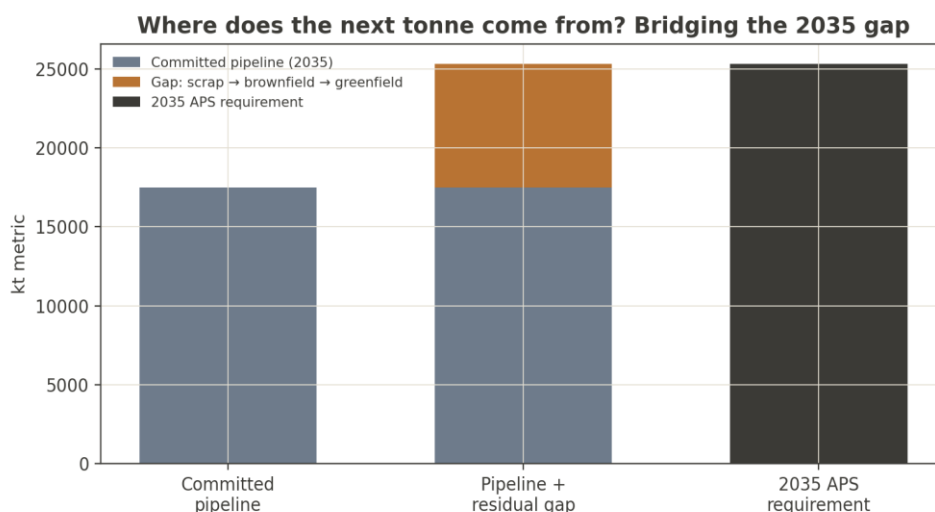
STRATEGIC TAKEAWAY

Conclusively, it is easier to understand that a single copper-price view dispenses inadequate strategic guidance for any of these six groups. Each faces a different mechanism, and each should size its response to that mechanism rather than to a shared price assumption. Having assigned a mechanism to each sector, the report now brings the framework to its most direct question: in what order, and from where, does the next tonne arrive?

13. Where Does the Next Tonne Come From?

Ordering in terms of elasticity, the next tonne follows this chronology: First from scrap, fastest to respond to price but capped by the stock of end-of-life metal followed by committed mine expansions in the existing producer base comprising Chile, Peru, the DRC. Only after that does it come from greenfield mines, which carry multi-year permitting, water and community risk. The constraint that most often stops the tonne is not ore but concentrate reaching a smelter, given how concentrated refining has become. The implication is that raising ambition toward net zero alone does not alter this ordering rather it lengthens the queue, demanding more of each source at once.

Figure 17. Where does the next tonne come from?



Strategic caption: The committed pipeline covers roughly 69% of the mid-decade requirement; the residual ~7.8 Mt/yr gap must be filled, in order of speed, by scrap, then brownfield expansion, then greenfield capacity.

Why this matters: This is the defining visual of the report: it converts the central question — where does the next tonne come from — into a single quantified bridge. The ordering shown is a qualitative elasticity ranking, not an independently disaggregated forecast of how the gap splits across sources.

The sequencing in Figure 17 is also a sequencing of policy leverage, developed further in Table 5 below. Scrap-collection policy acts on a horizon of months to a few years. The permitting reform for committed brownfield expansions acts on a multi-year horizon within existing footprints. Meanwhile, greenfield permitting reform which is the slowest and highest-uncertainty lever, is also the one most exposed to resource-nationalism risk in Chile, Peru and the DRC. A government or investor with a short time horizon has only one lever available; one with a 2035 horizon has all three, but only if greenfield permitting decisions are made now.

Source: Committed-pipeline capacity vs primary-supply requirement, with elasticity ranking overlaid.

Table 5. Relative Responsiveness of Incremental Copper Supply

Source	Speed to respond	Scale available	Execution risk
Scrap (old, post-consumer)	Fastest — responds to price within a cycle	Capped by end-of-life metal stock; global secondary share of demand ~17% rising to ~27.5%	Low, but volume-limited
Committed brownfield expansions	Medium — within existing permits and infrastructure	Bounded by the committed pipeline itself, ~17.5 Mt	Moderate - execution and cost risk on already-approved projects

Source	Speed to respond	Scale available	Execution risk
Greenfield mines	Slowest — multi-year permitting, water and community processes	Largest theoretical pool but least certain to convert to production on schedule	Highest - permitting and water-stress exposure concentrated in Chile, Peru, DRC

Table 5 — strategic interpretation: a strategy that treats these three sources as equally available mistakes elasticity for magnitude — sizing, sequencing and hedging decisions should follow the ranking, not the total.

STRATEGIC TAKEAWAY

The next tonne is not a single decision rather a sequence. Scrap buys the first two to three years, committed brownfield expansion buys the next several, and greenfield capacity is what determines whether the gap closes by the mid-2030s or persists beyond it.

INVESTOR PERSPECTIVE

What Sophisticated Investors Should Monitor

Institutional investors should look beyond the daily copper price to a set of leading indicators that typically move before price fully reflects a changing market. These include committed-pipeline conversion, treatment and refining charges (TC/RC), refining utilization, secondary-supply growth, permitting timelines, water-stress developments in the dominant producing countries, China's refining concentration, freight costs, marine insurance conditions, shipping disruptions, energy prices, strategic stockpile levels and the trajectory of climate-policy ambition.

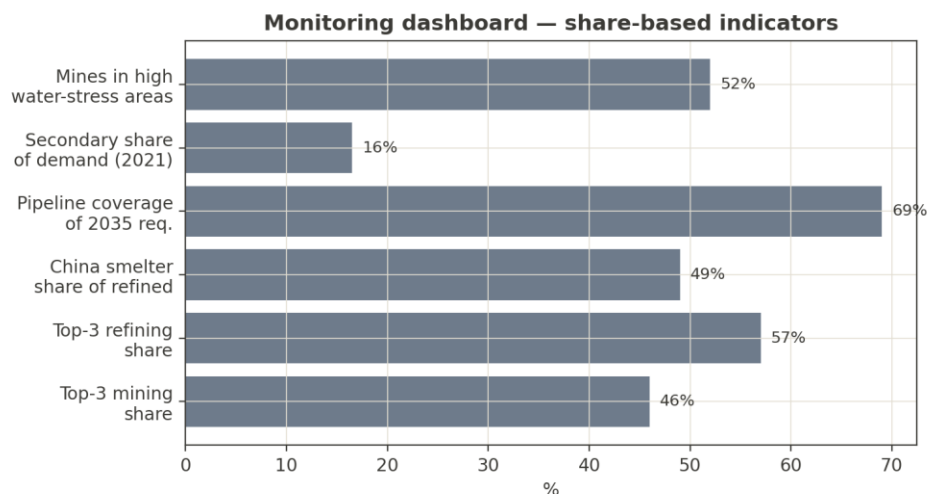
Each of these is a leading, not a lagging signal and explicates the mechanism through which a physical shortfall becomes visible before it reaches the price tape.

Strategic relevance: The physical tonne should be monitored before the price tape.

14. Monitoring Dashboard

A ten-indicator tracking framework presented in this report and designed to be refreshed each quarter as production, scenario and reserve data update.

Figure 18. What should the market watch every quarter?



Strategic caption: Committed-pipeline coverage of the mid-decade requirement (~69%) and refining concentration (~57%) are the two indicators most directly tied to this report's central finding.

Why this matters: This chart should be the first page a reader opens each quarter — a shift of a few points in pipeline coverage or refining concentration is the earliest signal that the gap is closing or widening.

Source: Composite of indicators listed in Table 6.

Table 6. Ten-indicator monitoring set

#	Indicator	Current reading
1	World mine production (Mt, latest)	21.9
2	Top-3 mining concentration share	49.9%
3	Top-3 refining concentration share	61%
4	China smelter share of world refined output	49%
5	Committed-pipeline coverage of mid-decade requirement	69%
6	Implied mid-decade pipeline shortfall (Mt/yr)	7.8
7	Secondary share of demand, most recent year	16.5%
8	Electrification premium, 2030 (Mt)	6.6
9	Net-zero vs pledges-based demand delta, 2030 (Mt)	1.9
10	Mines in high water-stress areas	52%

Table 6 — Strategic interpretation: indicators 5 and 6 are the leading signal for this report's central finding; indicators 2, 3 and 4 are the leading signal for downstream concentration risk; indicator 10 is the leading signal for operational disruption risk.

STRATEGIC TAKEAWAY

This report signifies that a quarterly reading of this dashboard, not a daily reading of the copper price, should be the ideal pace for strategic thinking.

15. What Is the Market Misunderstanding?

The market is not mispricing copper's near-term availability. Supply is adequate today, and our figures confirm it. The more consequential misunderstanding is about where attention should be directed. Market commentary has tended to frame the copper story around demand with respect to EV adoption, China and the pace of electrification. The framework underpinning this report depicts that demand growth, while real and structurally significant, is not the binding constraint. The binding constraint is how much of the forward requirement is already covered by committed capacity. On that specific measure, the market is short by design, not by surprise: about 31% of the mid-decade requirement has no committed project behind it yet.

A second, related misreading is treating recycling as a solution rather than a lever. The near-doubling of secondary supply by 2040, is the fastest-scaling part of this system, but even doubled, it does not close the primary-supply requirement, which plateaus near 25 Mt regardless. Recycling should be pursued aggressively and demonstrated as a means of narrowing the gap, not eliminating it.

A third misreading treats mining and refining concentration as a single risk, where our report presents evidence (Section 9, Table 2) that shows them to be materially different in degree. So, the top-3 mining concentration of roughly 50% is moderate by the standards of many commodities, while top-3 refining concentration around 61%, anchored by a single country smelting on a fraction of its own mined output, is a distinct and arguably more urgent policy problem. Industrial policy, critical-minerals strategy and supply-chain resilience planning that treat 'copper concentration' as one number rather than two are working from an incomplete picture of where the system's actual single points of failure sit.

Finally, the last misreading is that copper being priced and analyzed through historical cyclical and macroeconomic frameworks. In other words, the 'Dr. Copper' convention of reading the metal as a proxy for global growth. That framework is becoming a progressively less complete description of what moves the market. Copper's economics are increasingly shaped by structural, policy-linked forces that do not revert with the business cycle, such as, the energy transition, AI and data-centre buildout, electrification, grid expansion, industrial policy, critical-minerals strategy, geopolitical fragmentation and the resilience of global logistics networks. These forces layer a structural, capacity-constrained dimension onto copper's traditional cyclicity rather than replacing it.

What should readers monitor over the next several quarters?

- The pace at which announced copper projects convert to committed status. This is the single indicator most directly tied to whether the 31% pipeline shortfall narrows or widens.
- Secondary supply's share of total demand, and whether it is tracking ahead of or behind the pledges-based trajectory.
- Any shift in climate-policy ambition from a pledges-based toward a net-zero trajectory, which raises the primary-supply requirement independent of the pipeline itself.
- Refining-capacity announcements outside the current dominant hub, as a signal of whether downstream concentration risk is being actively addressed.
- Water-stress and permitting developments in Chile, Peru and the DRC, given their combined weight in both current production and reserve concentration.

What would invalidate this thesis?

- A material acceleration in committed-project conversion that closes the pipeline gap faster than the mid-decade horizon this report anchors on.
- A durable upside surprise in secondary-supply scaling that lifts its share of demand well above the 27.5% pledges-based milestone, materially easing the primary-supply requirement.
- A sustained retreat from climate-policy ambition that removes the electrification premium from the demand base entirely, rather than merely slowing its growth.

None of this argues that price is irrelevant. It is the mechanism through which scarcity, once physical, is ultimately rationed. It argues that price is a lagging signal of a condition this report's indicators can identify years in advance. A reader who tracks the copper price alone will learn that a shortfall has arrived only after it has already begun to bind. A reader who tracks committed-pipeline coverage, refining concentration and secondary-supply share, will see the same shortfall taking shape years before it reaches the price tape, with enough lead time for the permitting, financing and stockpiling decisions. That is the single lasting takeaway this report is written to leave behind: watch the physical tonne, not the price.

About This Framework

The Global Copper Balances & Physical-Tonne Framework is an original analytical framework developed using publicly available market data to explain copper supply through mine production, secondary recovery, committed capacity and processing constraints.

Sources

International Copper Study Group, world and country-level mine production and smelter/refinery concentration statistics; United States Geological Survey, Mineral Commodity Summaries and recycling statistics; International Energy Agency, Critical Minerals Outlook, Electricity Grids, and Global EV Outlook publications.

Disclosure

This report is for informational and institutional research purposes only and does not constitute investment advice. Figures are drawn from the sources listed above; scenario paths are direct extensions of published institutional milestones and do not incorporate price-elasticity feedback.

About the Author

Sidrah Azmat is an equity research professional with experience in financial modelling, valuation, capital allocation and energy market analysis. Her work focuses on integrating financial modelling with macroeconomic, commodity and industry research to develop investment frameworks for institutional investors, corporate strategy teams and policymakers. She regularly publishes research on energy markets, infrastructure, valuation and capital allocation.

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